
1 Bayes Theorem

1.1 Concepts

1. We use **Bayes theorem** when we want to find the probability of A given B but we are told the opposite probability, the probability of B given A . There are several forms of Bayes Theorem as follows:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|\bar{A})P(\bar{A})} = \frac{1}{1 + \frac{P(B|\bar{A})P(\bar{A})}{P(B|A)P(A)}}.$$

In order to discern which form to use, look at the information you are given. If you are told $P(B|A)$ as well as $P(B|\bar{A})$, use the latter two methods but if you are only told $P(B)$, then use the first form.

We say that two events A, B are **independent** if $P(A \cap B) = P(A)P(B)$.

1.2 Examples

2. There are 10 red and 10 blue balls in a bag. Someone randomly picks out a ball and then places it back and puts 10 more balls of that color into the bag. Then you draw a ball. What is the probability that the 10 balls added were red, given that you drew out a red ball?

Solution: We use Bayes Theorem to get

$$\begin{aligned} P(AddRed|DrawRed) &= \frac{1}{1 + \frac{P(DrawRed|AddBlue)P(AddBlue)}{P(DrawRed|AddRed)P(AddRed)}} \\ &= \frac{1}{1 + \frac{10/30 \cdot 1/2}{20/30 \cdot 1/2}} = \frac{1}{1 + 1/2} = \frac{2}{3}. \end{aligned}$$

3. Out of those brought to court, there are 60% which are actually guilty. Of those that are guilty, 95% of them are convicted. But there are 1% of innocent people who get falsely convicted. What is the probability that you are actually innocent given that you are convicted?

Solution: We use Bayes rule which tells us that $P(\text{Innocent}|\text{Convicted})$

$$\begin{aligned} &= \frac{P(\text{Convicted}|\text{Innocent})P(\text{Innocent})}{P(\text{Convicted}|\text{Innocent})P(\text{Innocent}) + P(\text{Convicted}|\text{Guilty})P(\text{Guilty})} \\ &= \frac{0.01 \cdot 0.4}{0.01 \cdot 0.4 + 0.95 \cdot 0.6} \approx 0.7\% \end{aligned}$$

1.3 Problems

4. True **FALSE** We can always use the formula $P(A|B) = \frac{1}{1 + \frac{P(B|A)P(A)}{P(B|A)P(A)}}$.

Solution: This is undefined if $P(B) = 0$.

5. I have two boxes of apples and oranges. In box 1, there are 5 oranges and 6 apples, in box 2 there are 6 oranges and 5 apples. I randomly pick a box and then in this box randomly pick a fruit. What is the probability that I picked box 1 given that I picked an orange?

Solution: We have that

$$\begin{aligned} P(\text{Box1}|\text{Orange}) &= \frac{P(\text{Box1} \cap \text{Orange})}{P(\text{Orange})} \\ &= \frac{P(\text{Orange}|\text{Box1})P(\text{Box1})}{P(\text{Orange}|\text{Box1})P(\text{Box1}) + P(\text{Orange}|\text{Box2})P(\text{Box2})} = \frac{5/11 \cdot 1/2}{5/11 \cdot 1/2 + 6/11 \cdot 1/2} = \frac{5}{11}. \end{aligned}$$

6. An exam has a 99% chance of testing positive if you have the disease and 1% chance of testing positive if you do not have the disease. Give that 0.5% of people have this disease, what is the probability that you have the disease given that you tested positive?

Solution:

$$\begin{aligned} P(\text{Sick}|\text{Positive}) &= \frac{P(\text{Positive}|\text{Sick})P(\text{Sick})}{P(\text{Positive}|\text{Sick})P(\text{Sick}) + P(\text{Positive}|\text{Notsick})P(\text{Notsick})} \\ &= \frac{0.99 \cdot 0.005}{0.99 \cdot 0.005 + 0.01 \cdot 0.995} \approx 0.33. \end{aligned}$$

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7. About $2/3$ of drivers use their cell phone while driving. Suppose that you are 5 times more likely to get into an accident if you text and drive, and if you don't use your cell phone, you have a 1% chance of getting into an accident. What is the probability that someone was texting given that they got into an accident?

Solution: We have that

$$\begin{aligned} P(\text{Text}|\text{Accident}) &= \frac{P(\text{Accident}|\text{Text})P(\text{Text})}{P(\text{Accident}|\text{Text})P(\text{Text}) + P(\text{Accident}|\text{NoText})P(\text{NoText})} \\ &= \frac{0.05 \cdot 2/3}{0.05 \cdot 2/3 + 0.01 \cdot 1/3} = \frac{10}{11}. \end{aligned}$$

2 Review

8. How many ways can 10 boys be paired up with 10 girls so that each boy is paired up with one girl.

Solution: The first boy has 10 choices, the second now has 9, and so on so $10 \cdot 9 \cdots 1 = 10!$.

9. How many ways can you arrange 10 marbles in a row if 4 are red, 3 are blue, and 3 are green (marbles of the same color are identical)?

Solution:

$$\binom{10}{4} \binom{6}{3} = \frac{10!}{4!3!3!}.$$

10. Prove that $\binom{n}{r} = \binom{n}{n-r}$ in two different ways.

Solution: We can first prove it algebraically by

$$\binom{n}{r} = \frac{n!}{r!(n-r)!} = \binom{n}{n-r}.$$

Then, to prove it combinatorially, the left side is the number of ways to choose a team of r people out of n and this is the same as choosing who is NOT on the team ($n-r$ people).

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11. Zvezda and Ramanujan play a game. They roll 4 6 sided die. If at least one 6 is rolled, then Zvezda wins. What is the probability that she wins?

Solution: The total number of ways the game can go is 6^4 . She wins if at least one 6 is rolled, which is 6^4 minus the number of ways no 6's are rolled. So, she wins with probability

$$1 - \frac{5^4}{6^4}.$$

12. How many solutions are there to

$$x_1 + x_2 + x_3 + x_4 + x_5 = 100$$

if $x_1 \geq 4, x_2 \geq 8, x_3 \geq 2, x_4 \geq 3, x_5 \geq 0$?

Solution: First we subtract to get rid of the restrictions to get $100 - 4 - 8 - 2 - 3 - 0 = 83$ balls and 5 boxes. So the answer is

$$\binom{83 + 5 - 1}{83} = \binom{87}{83}.$$

13. How many subsets of $\{1, 2, \dots, n\}$ contain at least one of 1 and 2.

Solution: We use PIE. There are 2^{n-1} subsets that contain 1, 2^{n-1} that contain 2, and 2^{n-2} that contain both. So the answer is

$$2^{n-1} + 2^{n-1} - 2^{n-2} = 2^n - 2^{n-2}.$$

14. Let $\{a_n\}_{n \geq 0}$ be the sequence defined by $a_0 = 1$ and $a_{n+1} = 4a_n + 1$. Prove that $a_n = \frac{4^{n+1}-1}{3}$ for all $n \geq 0$.

Solution: By induction on n .

Base Case ($n = 0$): Note $a_0 = 1 = \frac{4^1-1}{3}$.

Inductive Hypothesis: Assume $a_n = \frac{4^{n+1}-1}{3}$ for some $n \geq 0$. Inductive Step: Then $a_{n+1} = 4a_n + 1 = 4 \frac{4^{n+1}-1}{3} + 1 = \frac{4^{n+2}-4+3}{3} = \frac{4^{n+2}-1}{3}$.

Thus by induction $a_n = \frac{4^{n+1}-1}{3}$ for all $n \geq 0$.

15. Prove that if you select $n + 1$ distinct numbers from 1 to $2n$, then at least two of the numbers sum to $2n + 1$.

Solution: We pair them up as $(1, 2n), (2, 2n - 1), \dots, (n, n + 1)$. There are n pairs so by pigeonhole, we must select two in the same pair which means we have two that sum to $2n + 1$.

16. (Challenge) How many ways are there to sit 3 males and 7 females at a circular table so that no two males sit next to each other? (HINT: First try to do this problem when we only care about the order of M and F)

Solution: First let's assume that the males are all identical and females are identical and we just consider order of males and females. We pair up the males with a female so we need to seat 3 pairs of MF and 4 remaining 4 Fs. There are $\frac{7!}{3!4!} \cdot \frac{1}{7}$ ways to do this (we divide by 7 because rotating the table gives the same seating arrangement). Finally, once we determine the ordering of males and females, we multiply by $3!$ for the ways the three males to sit and $7!$ for the females to get

$$\frac{7!}{3!4!} \frac{3!7!}{7} = \frac{7!6!}{4!}.$$